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Sample Question Set - Unit 5

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO5
Paper: Sample Multiple Choice Questions
Maximum Marks: 40 x 1 = 40

Unit: 5
Unit Title: Model Evaluation and Improvement
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the regression metric that averages the absolute differences between the predictions and the actual values.

A) Mean absolute error
B) Mean squared error
C) Root mean squared error
D) R squared

Q2. Identify the entry of a confusion matrix that counts the positive cases correctly identified.

A) False positives
B) True positives
C) True negatives
D) False negatives

Q3. State what the F1 score combines.

A) Accuracy and recall, as their arithmetic mean
B) Precision and accuracy, as their product
C) Precision and recall, as their harmonic mean
D) Recall and specificity, as their difference

Q4. Name the regression metric obtained by taking the square root of the mean squared error.

A) Mean absolute error
B) R squared
C) Adjusted R squared
D) Root mean squared error

Q5. Identify the two quantities plotted against one another on a ROC curve.

A) Precision against recall
B) True positive rate against false positive rate
C) Accuracy against the threshold
D) Training error against test error

Q6. State what a global explanation of a model describes.

A) Why one particular record received its prediction
B) How long the model takes to make a prediction
C) How much memory the model needs to run
D) How the model behaves across the whole population

Q7. Explain why the root mean squared error is expressed in the same units as the target.

A) The square root undoes the squaring of the errors
B) The errors are divided by the number of records
C) The metric is always scaled between zero and one
D) The metric ignores the sign of every error

Q8. Distinguish precision from recall by the quantity each is measured over.

A) Recall is over the predicted positives, precision over the actual
B) Both of the two are measured over the predicted positives
C) Precision is over the predicted positives, recall over the actual positives
D) Both of the two are measured over the actual positives

Q9. Describe what an R squared of exactly zero indicates about a regression model.

A) It predicts every value perfectly correctly
B) It has an average error of exactly zero units
C) It does no better than always predicting the mean
D) It cannot be applied to this kind of data

Q10. Interpret a ROC-AUC of 0.5 reported for a classifier.

A) It performs no better than random guessing
B) It performs perfectly on every record
C) It has an accuracy of exactly 50 per cent
D) It always predicts the positive class

Q11. Predict the effect on precision of raising the decision threshold from 0.5 to 0.8.

A) Precision usually falls while recall rises
B) Both precision and recall usually rise
C) Neither precision nor recall is affected
D) Precision usually rises while recall falls

Q12. Recognize the situation described: a fraud model raises very few false alarms but catches only a small share of the frauds.

A) High recall with low precision
B) High precision with low recall
C) High values of both of the two
D) Low values of both of the two

Q13. Explain why the mean squared error penalises a large error more heavily than the mean absolute error does.

A) Squaring removes the sign of the error
B) Squaring divides the error by the record count
C) Squaring converts the error into a percentage
D) Squaring grows faster than the error itself

Q14. Summarize what cross validation gives that a single train-test split does not.

A) A model that is trained on far more data
B) A guarantee that the model will not overfit
C) An estimate that does not depend on one lucky split
D) A final performance figure for the report

Q15. Distinguish grid search from random search by the way each chooses the values it tries.

A) Random tries every combination, grid samples them
B) Grid tries every combination, random samples them
C) Both of the two try every combination in turn
D) Both of the two sample the combinations at random

Q16. Explain why explainability matters to an applicant a model has rejected.

A) They cannot challenge a decision they cannot understand
B) They would otherwise be unable to reapply
C) The explanation makes the model more accurate
D) The explanation shortens the processing time

Q17. Predict what happens when the hyperparameters are tuned by repeatedly checking the test set.

A) The model becomes impossible to train
B) The test score stops being an honest estimate
C) The test set grows larger with each check
D) The hyperparameters cease to have any effect

Q18. Describe what a local explanation accounts for.

A) Why one particular record received the prediction it did
B) Which features matter across all the records
C) How long the model took to be trained
D) How many records were used in training

Q19. Explain why a single accuracy figure can hide poor performance on one class.

A) Accuracy is computed on the training data only
B) Accuracy cannot be computed for two classes
C) Accuracy always equals the recall of the model
D) A small class contributes little to the overall figure

Q20. Distinguish a false positive from a false negative in a medical screening test.

A) A false negative alarms a healthy person
B) Both of the two miss an ill person
C) A false positive alarms a healthy person, a false negative misses an ill one
D) Both of the two alarm a healthy person

Q21. Compute the recall of a model that records 30 true positives and 20 false negatives.

A) 0.6
B) 0.4

C) 0.75
D) 1.5

Q22. Compute the precision of a model that records 30 true positives and 10 false positives.

A) 0.25
B) 0.6
C) 0.75
D) 3.0

Q23. Compute the accuracy of a model recording 40 true positives, 50 true negatives, 5 false positives and 5 false negatives.

A) 0.8
B) 0.9
C) 0.45
D) 0.95

Q24. Solve the following: a model records a precision of 0.6 and a recall of 0.4. Determine its F1 score.

A) 0.50
B) 0.24
C) 1.00
D) 0.48

Q25. Compute the mean absolute error of a model whose three errors are 2, 3 and 5.

A) 10
B) 5
C) About 3.33
D) About 3.87

Q26. Compute the mean squared error of a model whose two errors are 1 and 3.

A) 2
B) 4
C) 10
D) 5

Q27. Solve the following: a model records a mean squared error of 49. Determine its root mean squared error.

A) 7
B) 24.5
C) 49
D) 2401

Q28. Apply 4-fold cross validation to a dataset of 800 rows and determine the rows placed in the validation partition of each fold.

A) 600
B) 200
C) 400
D) 4

Q29. Compare a grid search over 3 values of one setting and 4 of another with a random search of 6 trials, on the number of combinations each evaluates.

A) The grid evaluates 7, the random search 6
B) The grid evaluates 6, the random search 12
C) Both of the two evaluate exactly 12
D) The grid evaluates 12, the random search 6

Q30. Examine a classifier recording 99 per cent accuracy on a dataset in which 99 per cent of the records are negative, and determine the likely explanation.

A) It has learned the positive class extremely well
B) It predicts the negative class for every record
C) It has been trained for too many iterations
D) The accuracy has been computed incorrectly

Q31. Use a change of the decision threshold from 0.5 to 0.7 and determine the effect on recall.

A) Recall rises
B) Recall is unchanged
C) Recall falls
D) Recall becomes undefined

Q32. Apply the definition of R squared to a model whose residual sum of squares is 30 and whose total sum of squares is 150.

A) 0.8
B) 0.2
C) 5.0
D) 0.5

Q33. Apply 10-fold cross validation and determine how many times the model is fitted.

A) 10
B) 1
C) 9
D) 100

Q34. Examine a model scoring 0.90 in cross validation but only 0.65 on data collected some months later, and determine the likely cause.

A) The cross validation used too many folds
B) The model was trained for too few iterations
C) The metric was computed with the wrong formula
D) The incoming data has drifted from the training data

Q35. Apply the confusion matrix to a screening test in which 100 cases are genuinely positive and 70 of them are detected. Determine the false negatives.

A) 70
B) 30
C) 100
D) 0

Q36. Use ROC-AUC values of 0.85 and 0.62 to select between two candidate classifiers.

A) The model scoring 0.62
B) Neither, since both are unusable
C) The model scoring 0.85
D) Either, since the two are equivalent

Q37. Apply a grid search over one setting with 3 values and another with 5, evaluated by 5-fold cross validation, and determine the number of model fits.

A) 15
B) 25
C) 75
D) 8

Q38. Examine a regression reporting a mean absolute error of 4 alongside a root mean squared error of 12, and determine what it indicates.

A) Every prediction is wrong by about the same amount
B) A few predictions are badly wrong
C) The model predicts the mean for every record
D) The two metrics have been swapped by mistake

Q39. Use the ideas of Explainable Artificial Intelligence to determine what a person refused a loan is entitled to.

A) A global explanation of the model
B) The full training dataset used
C) The source code of the model
D) A local explanation of their own case

Q40. Relate the number of hyperparameter combinations to the cost of running a grid search.

A) The cost is the product of the value counts times the folds
B) The cost is the sum of the value counts
C) The cost does not depend on the number of folds
D) The cost falls as more settings are added

----- End of Sample Question Set -----